

CLIMATE CHANGE DATA ANALYSIS

INTRODUCTION

Climate change is a complex environmental issue that involves multiple interacting indicators. Examining them helps us understand how they interact with each other and how they affect the climate. Indicators that are studied in this analysis are temperature, CO₂ emissions, renewable energy, forest area, sea level, rainfall, extreme weather events, population etc. Understanding these indicators individually is useful but understanding their relationship gives us meaningful insight to discern any particular pattern.

This project analyzes climate change dataset containing country level observations spanning across multiple years. The analysis uses MySQL for data exploration and statistical calculations and Power BI for interactive data visualization.

The purpose of this analysis is to identify patterns, relationship and statistical association within the dataset, rather than establishing causal relationships. The findings are established through what is observed within the dataset.

RESEARCH QUESTIONS

The analysis is structured around the following questions:

Descriptive analysis

1. How do the environmental indicators in the dataset vary across countries and over time?

Pairwise relationships

2. What apparent relationships exist between:
 - Average temperature and CO₂ emissions
 - Renewable energy share and CO₂ emissions
 - Extreme weather events and sea level rise

Multivariable analysis

- Temperature, CO₂ emissions, and Renewable Energy
- Temperature, Sea level, and Extreme weather events

DATASET OVERVIEW

Dataset:

- The dataset contains 1000 rows
- There are 15 countries in the dataset ranging from 2000 – 2023
- The dataset has no NULL values.

Methodology:

- MySQL: Used for data exploration, statistical calculations and comparison.
- Power BI: Used to create interactive data visualization and explore trends and patterns.

ANALYSIS AND CALCULATION

Descriptive Analysis: Analyzing individual environmental variables across countries spanning over various timelines to identify any trend or pattern that can be observed.

1. Temperature:

- The temperature trend shows fluctuation throughout the observation period rather than a simple consistent increase or decrease.
- The temperature trend has been observed across countries over varying time periods.

2. CO₂ emissions per capita:

- The average CO₂ emissions per capita is different across countries with some country having increased emission while others have relatively lower emission
- The average CO₂ emissions per capita has been observed to be relatively stable over time.

3. Extreme Weather Events:

- 2013 was observed to have highest extreme weather events while 2020 was observed to have least extreme weather events.
- Mexico was observed to have least extreme weather events while France was observed to have highest.

4. Sea Level Rise:

- The average sea level rise has been observed to gradually increase over time in the past year with highest sea level rise was observed to be in 2020
- The country that has been recorded to have the highest average sea level rise is Russia.

Relationship Analysis: Examining the relationship between selected environmental variables to identify if there is any apparent relationship that can be observed.

1. Temperature and CO₂ emissions:

- The scatter plot shows substantial dispersion and a slight upward trend line.
- The Pearson correlation coefficient is $r = 0.012$, indicating an extremely weak positive relationship.

2. Renewable Energy and CO₂ emissions:

- The scatter plot shows substantial dispersion.
- The Pearson correlation coefficient is $r = -0.023$, indicating an extremely weak negative relationship.

3. Extreme Weather events and Sea level rise:

- The scatter plot shows substantial dispersion and downward trend line.
- The Pearson correlation coefficient is $r = 0.025$, indicating an extremely weak positive relationship.

No substantial relationship was observed among the selected variable pair across overall country-year observation. However, results can vary through individual country observations.

Multivariable Analysis: Extending relationship analysis by examining three variables simultaneously. To investigate whether the relationship between two variables changed when a third variable was statistically accounted for, we used partial correlation.

1. Temperature, CO₂ emissions and Renewable Energy:

- The relationship between average temperature, CO₂ emissions and renewable energy were observed to be very weak.

- This indicates that accounting for renewable energy, the relationship between temperature and CO₂ emissions still remained weak.
- This suggests that accounting for a third variable did not substantially change the relationship between temperature and CO₂ emissions

2. Temperature, Sea level, and Extreme weather events:

- The relationship between average temperature, sea level and extreme weather events were observed to be very weak.
- This indicates that accounting for temperature, the relationship between sea level and extreme weather events still remained weak.

The partial correlation was used to examine the statistical association within available observation. It showed that accounting for the third variable didn't substantially change the previously observed relationship in pairwise relations.

KEY FINDINGS

- Average temperature showed fluctuation over time rather than consistent increase or decrease.
- CO₂ emissions per capita varied across countries, while overall yearly average remained relatively stable.
- Sea level rise showed gradual increase over year. Russia recorded the highest sea level rise among the countries within this dataset.
- No substantial relationship was found among the examined pairs of variables through Pearson correlation coefficient.
- Accounting for a third variable in the pairwise relationship didn't change the previously observed results in the pairwise relationship.

CONCLUSION

This analysis examined environmental indicators across countries over time using descriptive analysis, pairwise relationship and multivariable analysis.

The selected pairwise relationship evaluated using Pearson correlation coefficient showed no substantial correlation with one another. Even after accounting for a third variable the selected pairs of variables did not show change in the observation.

Overall, the relationship of different variables with one another and observing them individually using descriptive and statistical analysis provided insights on the pattern and relationship of different environmental indicators. The findings represent association observed within the dataset.